Simple Linear Regression and Correlation

Statistical Analysis for Industrial Engineers 2

BY: ENGR. MELAN DYRENE N. ALMENDRAS, SO2, PMFC, CLSSYB

INDEPENDENT AND DEPENDENT VARIABLES

GAS PRICE

VEGETABLE/ RICE PRICE

NUMBER OF PATIENTS

NUMBER OF NEEDED DOCTORS

Independent Variable:

The independent variable is the factor that the researcher manipulates or controls in an experiment.

Dependent/Response Variable:

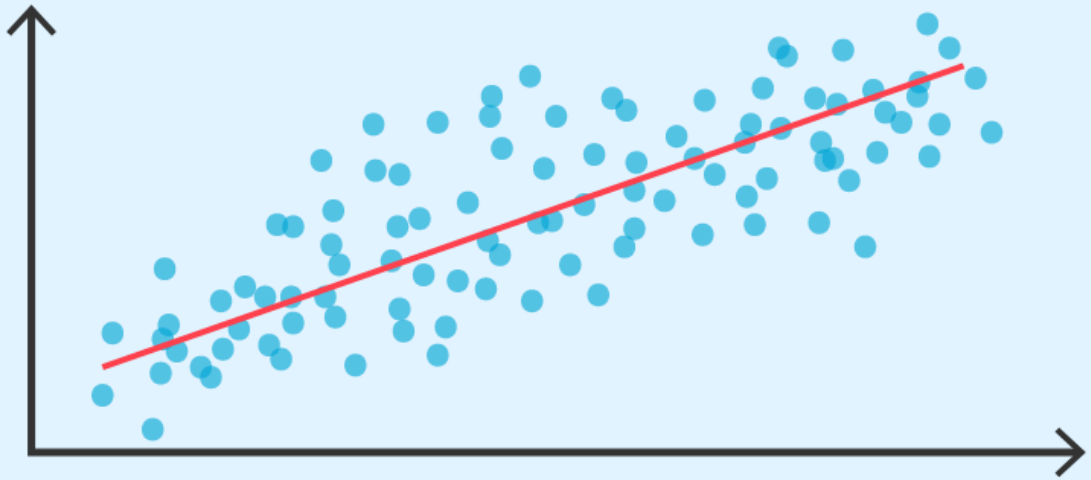
The dependent variable is the outcome or response that the researcher measures or observes during the experiment.

REGRESSION ANALYSIS is a reliable statistical method of estimating how a response variable depends on one or more predictors.

The concepts of regression analysis deals with;

- Finding the best relationship between Y and x,
- Quantifying the strength of that relationship, and
- Using methods that allow for prediction of the response values given values of the regressor x.

REGRESSION LINE is used to show the relationship of the variables. It may be interest of interest to develop a method of prediction, that is, a procedure for estimating the possible values of variables.



$Y = \alpha + \beta x + e$ is likely a reasonable approximation that may be operative within a confined range on x.

More often than not, the models that are simplifications of more complicated and unknown structures are linear in nature. These linear structures are simple and empirical in nature and are thus called **empirical models**.

Simple Linear Regression Model: $Y = \alpha + \beta x + e$

- Y = dependent variable
- x = independent variable
- α = y – intercept
- β = slope of the line
- e = error variance or residual variance

Important:

- ✓ The quantity **Y** is a random variable since e is random.
- ✓ The value **x** of the regressor variable is not random and, in fact, is measured with negligible error.
- ✓ The quantity ***e***, often called a ***random error*** or ***random disturbance,*** has constant variance. This portion of the assumptions is often called the *homogeneous variance assumption*.

- ***Fitted Regression Line***

  Then the estimated or fitted regression line is given by

  $$\hat{y} = b_0 + b_1 x \quad or \quad \hat{y} = a + bx,$$

  where $\hat{y}$ is the predicted or fitted value.

- ***True Regression Line***

  $$\mu_{Y/x} = \beta_0 + \beta_1 x$$

✓ ***Scatter Diagram*** – plot of values of x against values of y

✓ ***Residuals:*** $y_i - \mathrm{E}(Y_i) = y_i - (\alpha + \beta x_i) = \varepsilon_i$

Residual: Error in Fit

Given a set of regression data $[(x_i, y_i); i = 1,2,\ldots,n]$ and a fitted model, $fa = a + bx_i$, the *ith* residual e_i is given by

$$e_i = y_i - \hat{y}_i, \quad i = 1,2,\ldots,n.$$

- Small residuals are a sign of a ***good fit***.
- Another interesting relationship which is useful at times is the following: $y_i = (\alpha + \beta x_i) + e_i$

Least-Squares Approach

Formulas for the Regression Line

$$y' = A + Bx$$

$$A = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$B = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

Where a is the y' and b is the slope of the line

Example: Teacher Roi wants to see how the number of late of a students affects the student's final grade

Number of Late	5	10	4	0	7
Final Grade	78	70	89	94	76

x	y		
5	78		
10	70		
4	89		
0	94		
7	76		

Correlation: Estimating the Linear Relationship

Correlation

Correlation is a statistical measure that describes the extent to which two variables change together. In other words, it quantifies the degree to which there is a linear relationship between two sets of data.

The Pearson correlation coefficient, often denoted as "r," is a measure of the strength and direction of a linear relationship between two variables.

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[\, n\Sigma x^2 - (\Sigma x)^2 \,][\, n\Sigma y^2 - (\Sigma y)^2 \,]}}$$

Strength of Relationship

Size of Correlation	Interpretation
.90 to 1.00 (−.90 to −1.00)	Very high positive (negative) correlation
.70 to .90 (−.70 to −.90)	High positive (negative) correlation
.50 to .70 (−.50 to −.70)	Moderate positive (negative) correlation
.30 to .50 (−.30 to −.50)	Low positive (negative) correlation
.00 to .30 (.00 to −.30)	negligible correlation

Sample #1

A researcher is investigating the relationship between the number of hours students spend studying for an exam and their exam scores. The data collected from a sample of 10 students is in the table. Perform a simple linear regression analysis using the least squares method to determine the equation of the regression line. Interpret the slope and intercept in the context of the problem. Predict the exam score for a student who studies for 12 hours.

Hours Spent (X)	Exam Score (Y)
2	65
3	75
4	82
5	88
6	92
7	98
8	105
9	112
10	118
11	122

Sample #2

You would like to sell your used car. The car is about three years old. To get a good feeling for how much your car is worth, you collect prices of similar cars. You summarize the price and age in the table. The price is presented in thousand euros.

Price (k€)	Age (years)
29	1
25	2
21	2
18	3
15	4
15	5



Thank you for listening ☺